PEAD strategy suite -- comprehensive report

31 strategies, 1996-2013, $10M start, NAV-percentage Sharpe

1. League table

Every strategy that has been run, by asset class and Sharpe. `validated` strategies are candidates for use, `research` are explorations/variants, `reference` are kept as documented counter-examples.

name	family	status	ann. return	ann. vol	sharpe	max DD	calmar	Sharpe 96-01	Sharpe 02-07	Sharpe 08-13
s10_vol_rank_tilt	equity-volatility	research	7.1%	3.8%	1.81	-11.6%	0.61	2.94	1.89	0.78
s10_vol_rank_tilt_levered	equity-volatility	research	9.2%	5.2%	1.73	-15.9%	0.58	2.86	1.96	0.50
s9b_inverse_vol	equity-volatility	validated	6.7%	4.5%	1.47	-15.6%	0.43	2.40	1.63	0.59
s9b_inverse_vol_matched	equity-volatility	validated	7.3%	4.9%	1.45	-17.0%	0.43	2.39	1.66	0.52
s9c_inverse_vol_filtered	equity-volatility	research	6.5%	4.6%	1.41	-15.7%	0.42	2.26	1.57	0.60
s5_sector_neutral	equity-neutralized	validated	4.1%	3.0%	1.33	-10.8%	0.37	2.19	1.34	0.66
s6_neutral_levered	equity-neutralized	validated	6.5%	4.9%	1.30	-16.6%	0.39	2.07	1.52	0.47
s9a_vol_cell_filter	equity-volatility	research	6.3%	5.0%	1.23	-16.7%	0.38	1.90	1.44	0.48
s4_tilted_size_neutral	equity-neutralized	validated	5.2%	4.4%	1.18	-13.1%	0.40	1.74	1.31	0.60
s7_directional	equity-directional	reference	7.3%	6.6%	1.11	-15.5%	0.47	1.79	1.48	0.32
s6_beta_overlay_aggressive	equity-aggressive	research	15.0%	13.7%	1.09	-37.0%	0.41	1.39	1.73	0.21
rank_weighted	equity-baseline	validated	3.9%	3.9%	1.01	-14.0%	0.28	1.39	1.55	0.32
s5_sector_neutral_aggressive	equity-aggressive	research	12.1%	12.2%	1.00	-36.5%	0.33	1.53	1.71	-0.07
s6_neutral_levered_aggressive	equity-aggressive	research	12.1%	12.2%	1.00	-36.5%	0.33	1.53	1.71	-0.07
s7_directional_aggressive	equity-aggressive	research	11.8%	11.8%	1.00	-33.0%	0.36	1.64	1.65	0.13
s6_beta_overlay	equity-overlay	validated	10.1%	10.5%	0.97	-23.2%	0.43	1.30	1.14	0.48
extreme_decile	equity-baseline	validated	4.9%	5.3%	0.93	-18.5%	0.27	1.28	1.58	0.28
size_balanced	equity-baseline	validated	4.7%	5.3%	0.91	-17.5%	0.27	1.32	1.28	0.34
rank_weighted_aggressive	equity-aggressive	research	10.8%	12.4%	0.89	-38.8%	0.28	1.24	1.63	0.09
s4_tilted_size_neutral_aggressive	equity-aggressive	research	10.7%	12.3%	0.89	-40.0%	0.27	1.30	1.54	0.07
extreme_decile_aggressive	equity-aggressive	research	9.5%	12.2%	0.80	-36.2%	0.26	1.02	1.72	-0.01
size_balanced_aggressive	equity-aggressive	research	8.4%	12.3%	0.72	-37.5%	0.23	1.00	1.52	0.03
opt_kelly_cap20_legacy	options-kelly	reference	8.6%	2.2%	3.78	-10.3%	0.83	1.64	4.80	4.39
opt_kelly_cap30_legacy	options-kelly	reference	12.3%	3.3%	3.49	-17.4%	0.70	0.98	4.67	4.10
opt_kelly_cap40_legacy	options-kelly	reference	15.8%	4.5%	3.28	-23.9%	0.66	0.54	4.58	3.93
opt_kelly_cap20	options-kelly	validated	4.1%	2.3%	1.77	-13.8%	0.30	0.69	2.04	2.39
opt_kelly_cap30	options-kelly	validated	5.7%	3.6%	1.55	-22.3%	0.26	0.12	1.95	2.17

opt_kelly_cap20_conservative	options-kelly	research	2.3%	2.4%	0.99	-17.1%	0.14	0.32	0.99	1.60
hedged_blend_legacy_options	portfolio	reference	14.2%	4.2%	3.21	-9.9%	1.44	2.79	4.64	2.61
hedged_blend_s10	portfolio	research	12.4%	4.7%	2.50	-15.5%	0.80	2.92	2.84	2.10
hedged_blend	portfolio	validated	11.3%	4.8%	2.25	-16.7%	0.68	2.41	2.70	1.91



Each dot is one backtest. Options books and the hedged blend sit high-left (high return per unit of risk).

2. What each iteration changed

strategy	parent	d return	d vol	d sharpe	d max dd
extreme_decile_aggressive	extreme_decile	+4.62%	+6.92%	-0.12	-17.7%
hedged_blend	s9b_inverse_vol	+4.62%	+0.34%	+0.78	-1.1%
hedged_blend_legacy_options	hedged_blend	+2.92%	-0.65%	+0.96	+6.8%
hedged_blend_s10	hedged_blend	+1.10%	-0.10%	+0.25	+1.2%
opt_kelly_cap20_conservative	opt_kelly_cap20	-1.79%	+0.07%	-0.78	-3.3%
opt_kelly_cap30	opt_kelly_cap20	+1.57%	+1.31%	-0.22	-8.5%
opt_kelly_cap30_legacy	opt_kelly_cap20_legacy	+3.68%	+1.15%	-0.29	-7.1%
opt_kelly_cap40_legacy	opt_kelly_cap20_legacy	+7.21%	+2.32%	-0.50	-13.6%
rank_weighted_aggressive	rank_weighted	+6.95%	+8.53%	-0.11	-24.8%
s10_vol_rank_tilt	s9b_inverse_vol	+0.41%	-0.65%	+0.34	+4.0%
s10_vol_rank_tilt_levered	s10_vol_rank_tilt	+2.11%	+1.34%	-0.08	-4.3%
s4_tilted_size_neutral	rank_weighted	+1.37%	+0.55%	+0.18	+0.9%

s4_tilted_size_neutral_aggressive	s4_tilted_size_neutral	+5.45%	+7.84%	-0.29	-26.9%
s5_sector_neutral	s4_tilted_size_neutral	-1.19%	-1.39%	+0.15	+2.2%
s5_sector_neutral_aggressive	s5_sector_neutral	+8.09%	+9.18%	-0.33	-25.6%
s6_beta_overlay	s6_neutral_levered	+3.59%	+5.57%	-0.33	-6.6%
s6_beta_overlay_aggressive	s6_neutral_levered_aggressive	+2.86%	+1.49%	+0.09	-0.5%
s6_neutral_levered	s5_sector_neutral	+2.43%	+1.92%	-0.03	-5.8%
s6_neutral_levered_aggressive	s6_neutral_levered	+5.66%	+7.26%	-0.30	-19.8%
s7_directional	s6_neutral_levered	+0.80%	+1.62%	-0.19	+1.1%
s7_directional_aggressive	s7_directional	+4.48%	+5.29%	-0.11	-17.4%
s9a_vol_cell_filter	s6_neutral_levered	-0.23%	+0.11%	-0.07	-0.1%
s9b_inverse_vol	s6_neutral_levered	+0.22%	-0.45%	+0.17	+1.0%
s9b_inverse_vol_matched	s9b_inverse_vol	+0.60%	+0.45%	-0.02	-1.4%
s9c_inverse_vol_filtered	s6_neutral_levered	+0.05%	-0.39%	+0.12	+1.0%
size_balanced	extreme_decile	-0.17%	-0.07%	-0.02	+0.9%
size_balanced_aggressive	size_balanced	+3.71%	+7.00%	-0.18	-19.9%

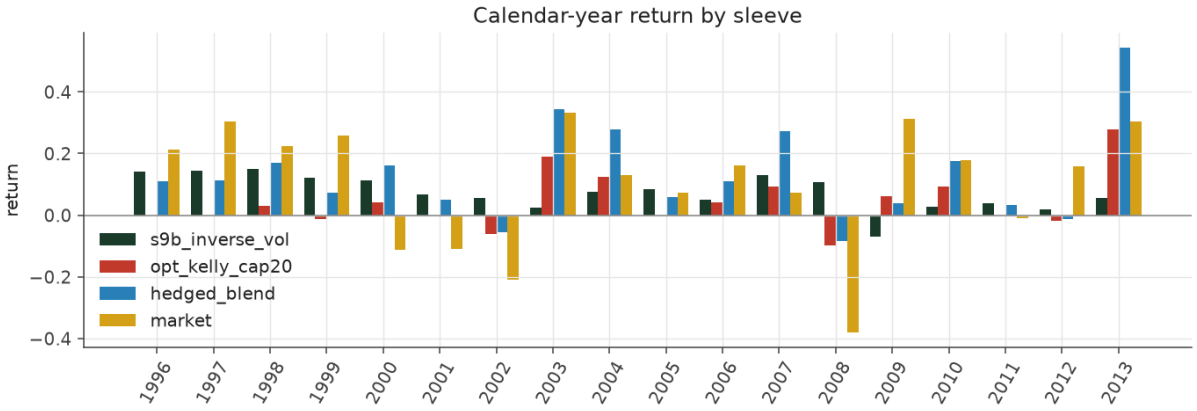
d_* columns are the change from the parent strategy. Reading down: neutralizing size and sector (S4->S5) and inverse-vol sizing (S6->S9b) buy Sharpe; raising the leverage cap buys return; the aggressive variants buy return at the cost of Sharpe because the liquidity cap starts to bind.

3. Stability across eras

name	Sharpe 96-01	Sharpe 02-07	Sharpe 08-13
extreme_decile	1.28	1.58	0.28
rank_weighted	1.39	1.55	0.32
s4_tilted_size_neutral	1.74	1.31	0.60
s5_sector_neutral	2.19	1.34	0.66
s6_neutral_levered	2.07	1.52	0.47
s7_directional	1.79	1.48	0.32
s9b_inverse_vol	2.40	1.63	0.59
s9a_vol_cell_filter	1.90	1.44	0.48
s10_vol_rank_tilt	2.94	1.89	0.78
s6_beta_overlay	1.30	1.14	0.48
opt_kelly_cap20	0.69	2.04	2.39
opt_kelly_cap30	0.12	1.95	2.17
opt_kelly_cap20_legacy	1.64	4.80	4.39
hedged_blend	2.41	2.70	1.91

year	s9b inverse vol	opt kelly cap20	hedged blend	market

1996	14.0%	0.0%	11.1%	21.2%
1997	14.4%	0.0%	11.4%	30.4%
1998	14.8%	2.9%	16.9%	22.3%
1999	12.0%	-1.2%	7.3%	25.7%
2000	11.3%	4.0%	16.1%	-11.3%
2001	6.8%	-0.3%	5.0%	-11.0%
2002	5.7%	-6.1%	-5.5%	-20.9%
2003	2.5%	18.8%	34.3%	33.1%
2004	7.6%	12.3%	27.7%	13.0%
2005	8.5%	-0.5%	5.9%	7.3%
2006	4.9%	4.2%	11.0%	16.2%
2007	12.9%	9.3%	27.1%	7.3%
2008	10.6%	-10.0%	-8.3%	-38.2%
2009	-6.9%	6.1%	3.9%	31.2%
2010	2.6%	9.2%	17.5%	17.7%
2011	4.0%	-	3.2%	-1.1%
2012	1.8%	-1.7%	-1.3%	15.8%
2013	5.5%	27.8%	54.4%	30.5%



Equity alpha is front-loaded (1996-2003); the options book is back-loaded (2003-2013) and lost in 2008 when equity gained.

The equity drift has decayed: Strategy 9b's return fell from ~14% a year in 1996-99 to ~2-5% in 2011-13, the usual signature of PEAD being arbitraged. The options book shows the opposite shape in this sample, which is part of why the two combine well -- and part of why its recent strength should not be extrapolated without a skeptical look.

4. Equity books: long leg vs. short leg

strategy	long pnl M	short pnl M	long excess M	short excess M	long bp per dollar	short bp per dollar	corr long short
extreme_decile	24.6	-5.7	10.9	6.1	275.1	-68.3	-0.68

rank_weighted	21.7	-7.1	7.3	6.7	248.5	-83.6	-0.79
s4_tilted_size_neutral	29.6	-8.1	13.2	8.8	297.5	-77.0	-0.78
s5_sector_neutral	20.0	-5.4	9.6	5.3	304.7	-80.7	-0.79
s6_neutral_levered	42.7	-11.8	15.9	15.9	253.4	-68.4	-0.81
s9b_inverse_vol	43.2	-10.8	15.9	17.2	253.2	-61.6	-0.81

• **In every equity strategy the short leg loses money in absolute terms** while the long leg earns several times the total. The headline is real but misleading: returns are measured against the cap-weighted market, and the average event stock beats that benchmark (small-cap tilt plus the earnings-announcement premium), flattering the long leg and penalizing the short leg by the same amount.

• Stripped of that common component, **the surprise signal earns the same on both legs** (columns `*_excess_M`): the shorts are pulling their weight, they are just paid in relative performance, not in absolute declines.

• The legs' daily P&L are ~-0.8 correlated: the short leg is a hedge. Dropping it for a long-only book would multiply the volatility and add market-relative factor exposure without adding alpha.

• The backtest charges no stock-borrow fee. The short leg is small-cap-heavy where borrow is expensive; each 1% a year of borrow costs ~0.6pp of Strategy 6's return (see the per-strategy report's borrow table).

• Alpha per dollar traded is ~7x higher in the smallest size quintile (~11x in the least liquid quintile) than in the largest, and the >90-day holding cells earn ~8x the alpha per dollar of the <=40-day cells. The book's capacity is therefore bounded by small illiquid names.

5. Options book: what the return is made of, and how much of it is real

The originally reported options result does not survive a careful look at the quotes. The original backtest marked a missing forward quote at the previous live quote. 22% of picks have no quote at 60 days and 93% of those are zero-bid options -- dead contracts nobody will buy -- which that convention never marked down. Corrected: 8.58% / Sharpe 3.78 as reported -> 4.12% / 1.77 (dead quotes at half-ask, plus a $0.05 minimum tick cost) -> 2.34% / 0.99 if dead options are worth nothing. A real, smaller edge remains.

• 98.2% of funded trades are calls, producing 96.5% of gross P&L -- across ALL deciles, including the worst surprises. The Kelly selector finds cheap OTM calls better than puts even on bad news.

• Win rate 39%, median trade -24%, mean 25%: a convex pay-off. The top 1% of trades are 65% of gross P&L.

• What survives the corrected marks is concentrated in calls struck at or just out of the money (1.0-1.2x spot) on low-volatility stocks. Deep in-the-money calls absorb about half the premium budget and earn ~5% on it; high-volatility stocks earn ~7%.

• Return on premium does not rise with SUE decile, so the edge is not the directional drift alone: it is mostly cheap post-earnings convexity on calm stocks, in a sample that was net bullish and whose profit is dominated by two years (2003 and 2013).

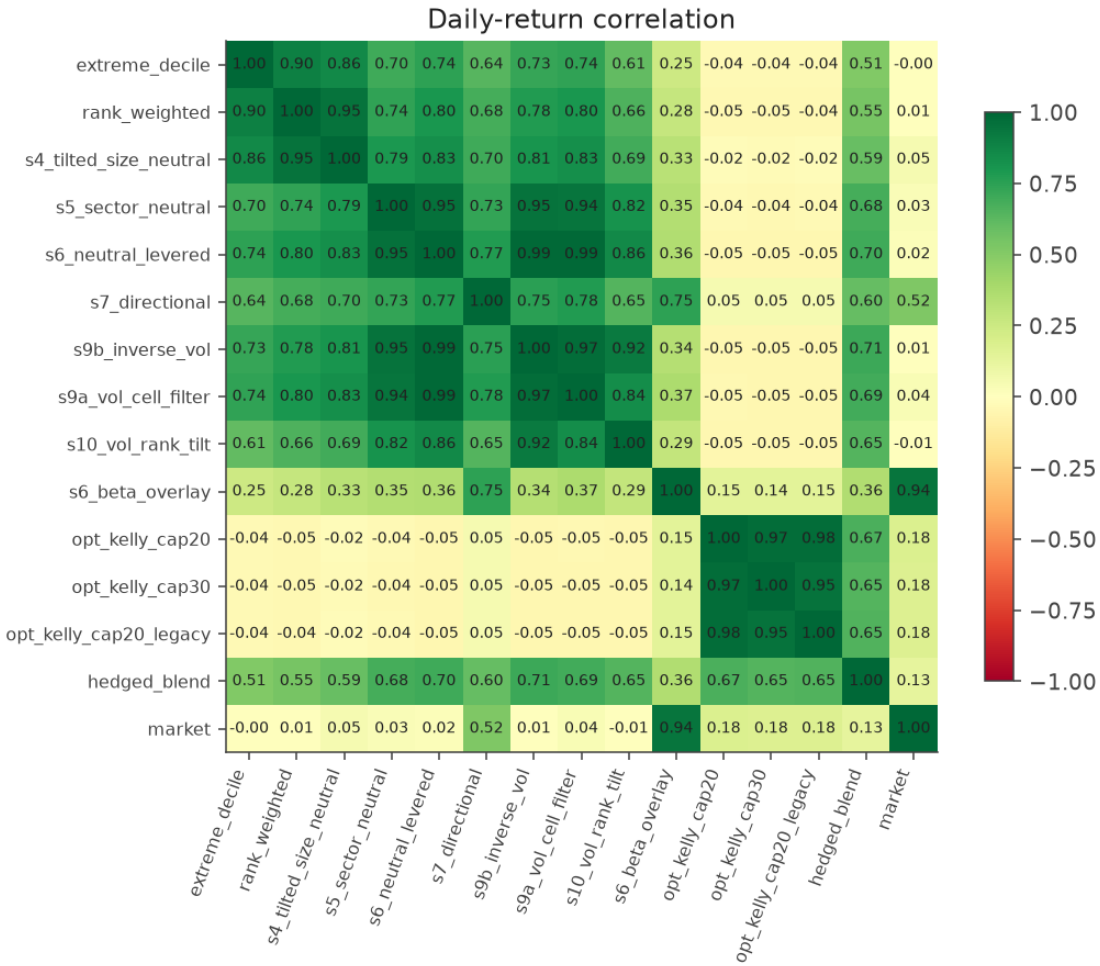
vol tercile	trades	premium M	gross M	net M	share of net	roi on premium
low vol	21,662	19.07	9.74	6.80	70.6%	51.1%
mid vol	21,661	16.81	3.68	2.17	22.5%	21.9%
high vol	21,662	25.63	1.86	0.66	6.8%	7.3%

Execution and marking sensitivity

case	ann. return	ann. vol	sharpe	max DD	trades
as registered	4.12%	2.30%	1.77	-13.8%	65,003
ORIGINAL convention: dead quotes carried forward, flat 3% cost	8.58%	2.18%	3.78	-10.3%	70,167
dead quotes at half-ask, flat 3% cost (no tick floor)	5.87%	2.29%	2.50	-12.7%	67,294
dead quotes worth zero (pessimistic bound)	2.34%	2.37%	0.99	-17.1%	62,581
+ $0.10/share minimum cost	1.98%	2.32%	0.86	-16.5%	62,288
skip contracts under $0.25	3.58%	1.94%	1.83	-12.9%	52,686
skip contracts under $0.50	2.79%	1.72%	1.61	-13.7%	43,993
flat 10% round-trip cost	2.75%	2.31%	1.18	-18.7%	62,809

Treat the corrected `opt_kelly_cap20` / `opt_kelly_cap30` as the options sleeve and the `_legacy` strategies as the historical headline only. Everything the hedged blend says about options is built on the corrected numbers.

6. How the strategies move together



Equity books cluster together; options are independent of them.

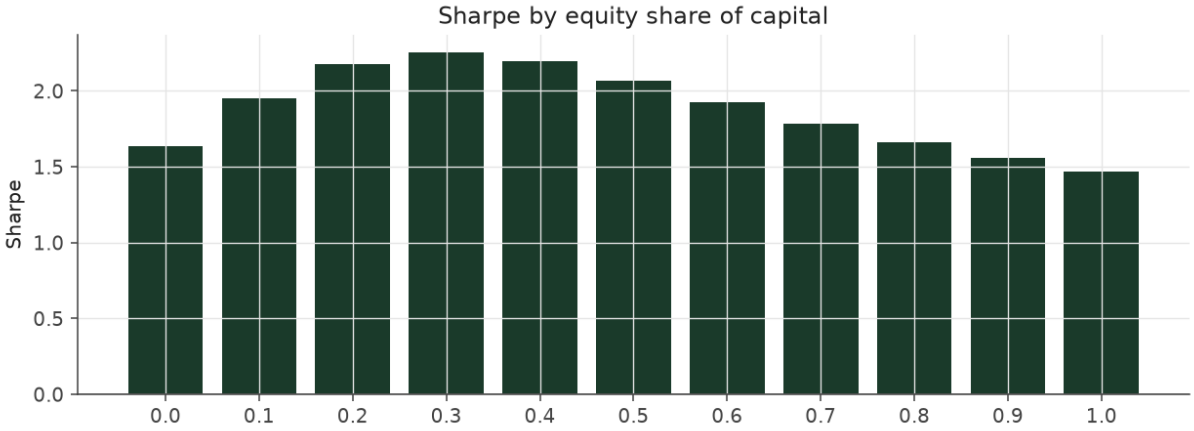
• All market-neutral equity variants are 0.7-0.99 correlated -- they are one bet expressed several ways, so combining them adds nothing.

• Equity vs. options correlation: -0.05. The options book's correlation to the market is 0.18; the market-neutral equity books' is ~0.

• `s6_beta_overlay` is 0.94 correlated with the market by construction; it answers 'beat the S&P', not 'diversify'.

7. Combining an equity book with an options book

Capital split: fraction of NAV in each sleeve (weights sum to 1)



Sharpe peaks around 15-25% equity: diversification, then the better sleeve dominates.

method	w equity	w options	ann. return	ann. vol	sharpe	max DD
equal	0.50	0.50	5.1%	2.4%	2.07	-6.3%
inverse_vol	0.32	0.68	4.6%	2.0%	2.25	-7.5%
max_sharpe (in-sample)	0.30	0.70	4.5%	2.0%	2.25	-7.8%
walk-forward max-Sharpe (out-of-sample)	0.49	0.51	5.0%	2.5%	1.99	-6.8%

Stacked on one NAV (weights are multiples of each sleeve's own sizing)

target vol	equity multiple	options multiple	ann. return	ann. vol	sharpe	max DD	worst day
3%	0.48	1.02	6.9%	3.0%	2.25	-11.1%	-1.6%
5%	0.81	1.70	11.8%	5.0%	2.25	-17.8%	-2.7%
8%	1.29	2.72	19.3%	8.0%	2.25	-27.1%	-4.3%

Equal-risk weights (each sleeve contributes the same stand-alone volatility). The 5% row is the registered `hedged_blend`. Larger multiples assume the extra options premium and equity gross exposure can actually be financed and traded at the backtested costs.

The hedge in action: stress years

year	s9b inverse vol	opt kelly cap20	hedged blend	market
2000	11.3%	4.0%	16.1%	-11.3%
2001	6.8%	-0.3%	5.0%	-11.0%

2002	5.7%	-6.1%	-5.5%	-20.9%
2008	10.6%	-10.0%	-8.3%	-38.2%
2009	-6.9%	6.1%	3.9%	31.2%

• The options book's worst year (2008, -10.0%) coincided with the equity book's +10.6%; in 2009 the signs flipped. That offset is the hedge -- it comes from low correlation across time, not from any offsetting position.

• The blend's market beta is 0.037: there is nothing left for an index hedge to remove. Layering a short-index overlay on top would only pay costs; a market-beta overlay (`s6_beta_overlay`) goes the other way, adding the equity risk premium on purpose.

• Everything here is in-sample except the walk-forward row. The caveats that apply to the sleeves apply to the blend: the options sleeve's costs, the equity sleeve's missing borrow fees, and no financing cost on the stacked multiples.

8. Conclusions and next steps

• Use `s9b_inverse_vol` as the equity alpha sleeve (validated, no fitted parameters, ahead of Strategy 6 in every era; `s10_vol_rank_tilt` is the stronger research candidate) and `opt_kelly_cap20` (or the cap30 variant) as the options sleeve -- a real but modest edge once dead quotes are marked down -- and blend by equal risk (`hedged_blend`).

• Do not add more market-neutral equity variants -- they are one bet. New equity ideas must change the signal or the holding rule, not the sizing.

• Options: the quote convention was the biggest error found in this project. Next: backtest with real bid/ask on entry and exit (the data has them), restrict to the contract region that survives (0.97-1.2x moneyness, low vol, premium >= $0.25), and test on a holdout -- 2003 and 2013 carry most of the profit.

• The measured weak spots to attack on the equity side: the era decay (alpha per dollar fell ~4x), the <=40-day holding cells that earn little after costs, and the mid deciles (D7, D9) that earn almost no excess per dollar.